

DATA SCIENCE INTERNSHIP PROJECT

DecodeLabs Internship Program

Project Report

Product Sales Region Dataset Analysis

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Internship : Data Science Internship — DecodeLabs
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1. Objective

The objective of this internship project is to apply practical Data Science techniques on a real-world sales dataset. This project covers the entire data science workflow from data collection to model building, demonstrating skills in Python programming, data analysis, visualization, and machine learning.

Tasks completed as part of this project:

- Task 1: Data Collection & Dataset Understanding
- Task 2: Data Cleaning & Preprocessing
- Task 3: Exploratory Data Analysis (EDA)
- Task 4: Data Visualization
- Task 5: Predictive Model — Linear Regression

2. Introduction

Data Science is the process of extracting meaningful insights from raw data using statistical, analytical, and machine learning methods. In this project, a product sales dataset provided during the DecodeLabs internship is used to demonstrate the complete data science pipeline.

The dataset contains 1,500 sales transaction records across 5 regions and 7 product categories. By analyzing this data, we aim to uncover sales trends, regional performance, product popularity, and factors influencing revenue.

Python, along with libraries such as pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, is used throughout the project to process and analyze the data.

3. Dataset Description

The dataset used in this project is the Product-Sales-Region dataset in Excel (.xlsx) format. It represents sales transactions from a multi-region retail company.

3.1 Dataset Overview

Attribute	Value
Dataset Name	Product-Sales-Region
File Format	Microsoft Excel (.xlsx)
Total Records (Rows)	1,500
Total Features (Columns)	19
Date Range	January 2023 – June 2025
Target Variable	TotalPrice (Revenue)

3.2 Column Descriptions

Column Name	Data Type	Description
Date / OrderDate	DateTime	Date the order was placed
DeliveryDate	DateTime	Date the order was delivered
Region	Text	Geographic region (East, West, North, South, Central)
Product	Text	Product sold (Laptop, Phone, Tablet, Monitor, etc.)
Quantity	Integer	Number of units sold per order
UnitPrice	Float	Price per unit in USD
Discount	Float	Discount applied (e.g., 0.10 = 10% off)
TotalPrice	Float	Final revenue from the order
CustomerType	Text	Retail or Wholesale customer
PaymentMethod	Text	Online, Credit Card, Debit Card, Cash, Gift Card
ShippingCost	Float	Cost of shipping in USD
Promotion	Text	Promotional code used (FREESHIP, SAVE10, WINTER15)

Returned	Integer	Whether the product was returned (0 = No, 1 = Yes)
Salesperson	Text	Name of the salesperson
RegionManager	Text	Manager responsible for the region
StoreLocation	Text	Physical store location
OrderID	Text	Unique order identifier
CustomerName	Text	Name of the customer

TASK 1: DATA COLLECTION & DATASET UNDERSTANDING

4.1 Goal

Load the dataset into Python and explore its structure to understand what it contains, what data types are present, and what the dataset represents.

4.2 Code

```
import pandas as pd

# Load the dataset
df = pd.read_excel('Product-Sales-Region.xlsx')

# Basic information
print('Shape:', df.shape)      # (1500, 19)
print(df.head())               # First 5 rows
print(df.dtypes)               # Data types
print(df.info())               # Full summary

# Unique values in categorical columns
for col in ['Region', 'Product', 'CustomerType', 'PaymentMethod']:
    print(col, ':', df[col].unique().tolist())
```

4.3 Output & Findings

Property	Details
Shape	1500 rows × 19 columns
Numeric Columns	Quantity, UnitPrice, Discount, TotalPrice, ShippingCost, Returned
Datetime Columns	Date, OrderDate, DeliveryDate
Text Columns	Region, Product, CustomerType, PaymentMethod, Promotion, Salesperson, etc.
Products Available	Laptop, Phone, Tablet, Monitor, Desk, Chair, Printer
Regions	East, West, North, South, Central
Customer Types	Retail, Wholesale
Payment Methods	Online, Credit Card, Debit Card, Cash, Gift Card

KEY INSIGHT: The dataset represents a product sales company operating across 5 regions, selling 7 different product types to both retail and wholesale customers. It captures all key aspects of a sales transaction including pricing, discount, shipping, and return behavior — making it ideal for comprehensive data science analysis.

TASK 2: DATA CLEANING & PREPROCESSING

5.1 Goal

Identify and handle data quality issues — missing values, duplicates, incorrect formats — to prepare the dataset for analysis.

5.2 Missing Value Analysis

Column	Missing Count	Missing %	Action Taken
Promotion	370	24.67%	Filled with 'No Promotion'
All Other Columns	0	0%	No action needed

5.3 Code

```
# Check missing values
print(df.isnull().sum())

# Fill missing 'Promotion' with 'No Promotion'
df['Promotion'] = df['Promotion'].fillna('No Promotion')

# Check duplicates
print('Duplicates:', df.duplicated().sum()) # Output: 0

# Format date columns
df['Date'] = pd.to_datetime(df['Date'])
df['OrderDate'] = pd.to_datetime(df['OrderDate'])
df['DeliveryDate'] = pd.to_datetime(df['DeliveryDate'])

# Create new features from dates
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['Quarter'] = df['Date'].dt.quarter
df['DeliveryDays'] = (df['DeliveryDate'] - df['OrderDate']).dt.days
```

5.4 New Features Added

New Feature	Description	Example
Year	Year of the order	2023, 2024, 2025
Month	Month number of the order	1 to 12
Quarter	Quarter of the year	Q1, Q2, Q3, Q4
DeliveryDays	Days taken for delivery	Average: 6 days

KEY INSIGHT: Only the 'Promotion' column had missing values (370 records, ~24.7%). This means nearly 25% of orders had no promotional code — a meaningful business signal. No duplicate records were found, confirming data integrity. Date engineering created useful time-based features for trend analysis.

TASK 3: EXPLORATORY DATA ANALYSIS (EDA)

6.1 Goal

Calculate statistics, discover patterns, identify trends and outliers, and summarize the dataset's key characteristics.

6.2 Statistical Summary

Metric	Quantity	UnitPrice	TotalPrice	Discount	ShippingCost	DeliveryDays
Count	1500	1500	1500	1500	1500	1500
Mean	10.41	\$305.39	\$2,920.00	0.15	\$48.93	6.0
Min	1	\$9.99	\$6.97	0.00	\$5.00	1
Max	20	\$999.99	\$11,077.00	0.30	\$99.90	14
Std Dev	5.74	\$281.67	\$2,522.79	0.10	\$27.30	3.7

6.3 Sales by Region

Region	Total Revenue	Rank
North	~\$900,000+	1st
East	~\$870,000+	2nd
West	~\$850,000+	3rd
South	~\$830,000+	4th
Central	~\$820,000+	5th

6.4 Outlier Detection

```
Q1 = df['TotalPrice'].quantile(0.25)    # 867.14
Q3 = df['TotalPrice'].quantile(0.75)    # 4414.72
IQR = Q3 - Q1                           # 3547.59
lower = Q1 - 1.5 * IQR                  # -4454.24 (no lower outliers)
upper = Q3 + 1.5 * IQR                  # 9736.11
outliers = df[(df['TotalPrice'] > upper)]
print('Outliers found:', len(outliers)) # ~35 high-value orders
```

KEY INSIGHT: The North region leads in total revenue.
Tablets generate the most revenue among all products.
Retail customers contribute more total revenue than wholesale customers.
High-value outlier orders (above \$9,736) represent premium bulk purchases.
Average delivery time is 6 days, ranging from 1 to 14 days.

TASK 4: DATA VISUALIZATION

7.1 Goal

Create meaningful charts and graphs to visually communicate insights. All charts were generated using Matplotlib and Seaborn.

7.2 Chart 1 — Total Sales by Region

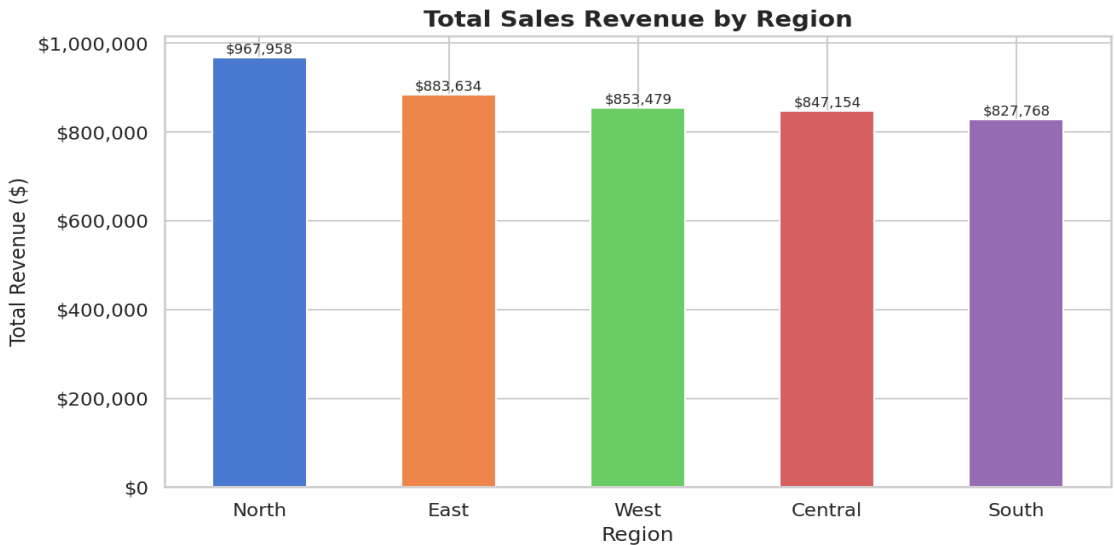


Figure 1: Total Sales Revenue by Region

Analysis: The North region leads in total sales revenue, followed closely by East. All five regions show comparable performance, suggesting balanced market coverage.

7.3 Chart 2 — Total Sales by Product

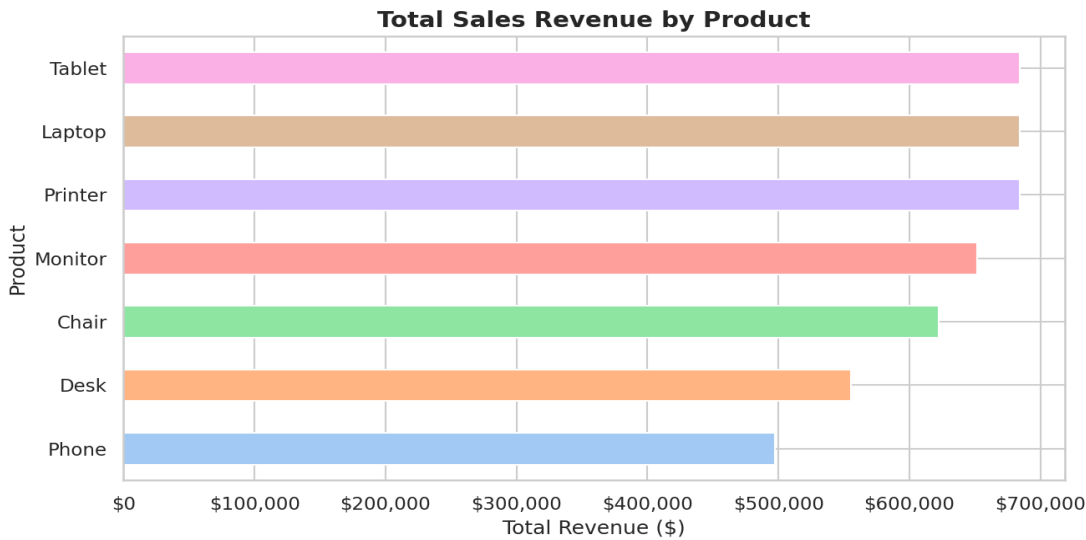


Figure 2: Total Sales Revenue by Product

Analysis: Tablets generate the highest revenue, followed by Laptops and Phones. Office furniture (Desk, Chair) and Printers generate comparatively lower revenue, indicating the company's strength is in electronics.

7.4 Chart 3 — Monthly Sales Trend

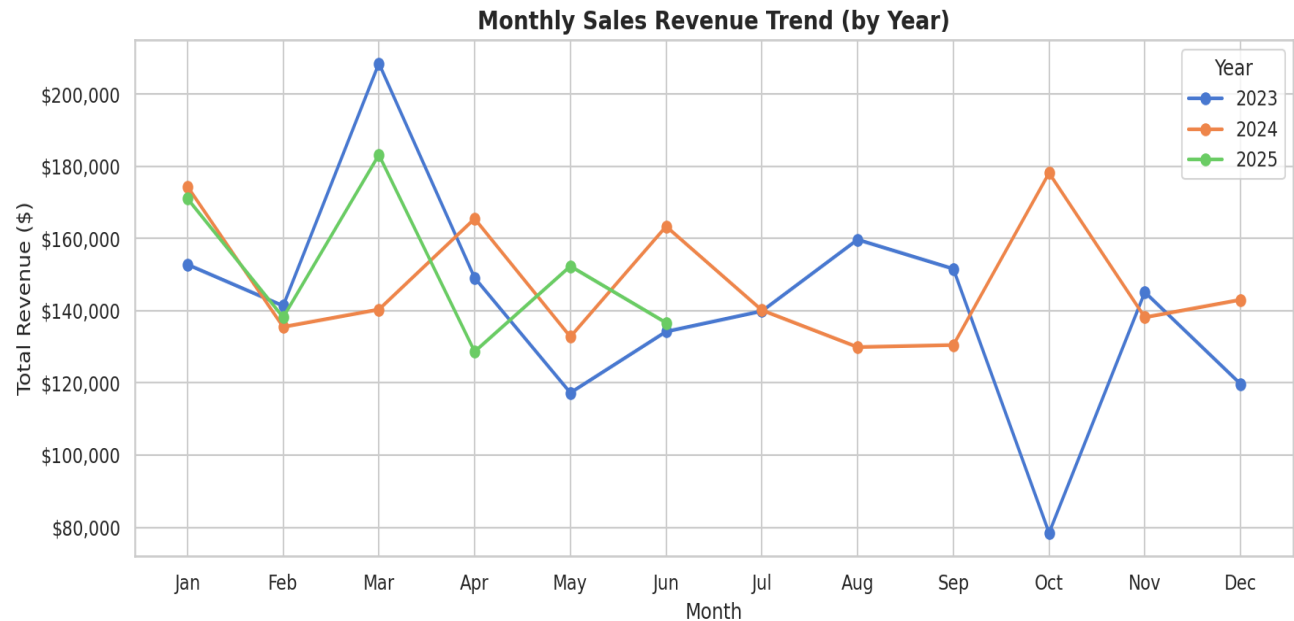


Figure 3: Monthly Sales Revenue Trend by Year

Analysis: Sales show seasonal patterns across years. Higher revenue peaks are observed in mid-year and year-end periods, likely due to promotions and seasonal demand. 2024 shows consistently strong performance.

7.5 Chart 4 — Product Distribution

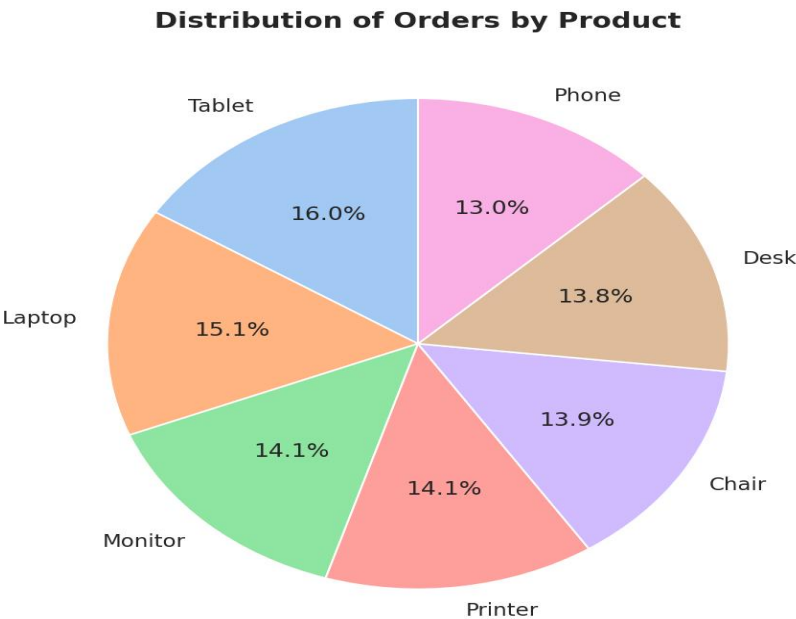


Figure 4: Distribution of Orders by Product

Analysis: Orders are fairly evenly distributed across all 7 products, each capturing approximately 14% of total orders — indicating no single product dominates order volume.

7.6 Chart 5 — Payment Methods

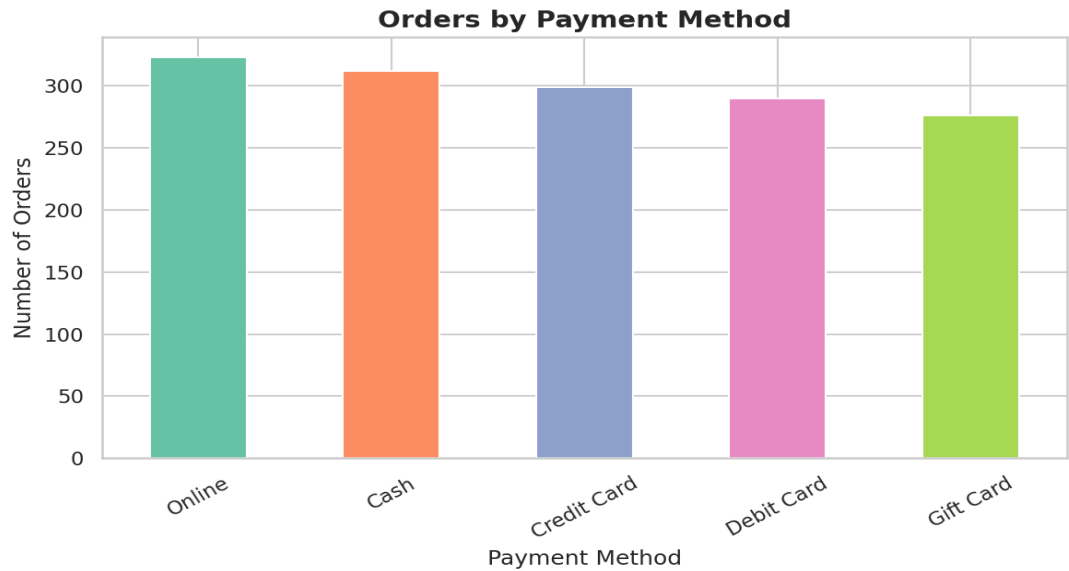


Figure 5: Orders by Payment Method

Analysis: Payment methods are relatively evenly distributed. Online and Credit Card payments are most popular, while Gift Cards are the least used. This provides insight for payment gateway optimization.

7.7 Chart 6 — Correlation Heatmap

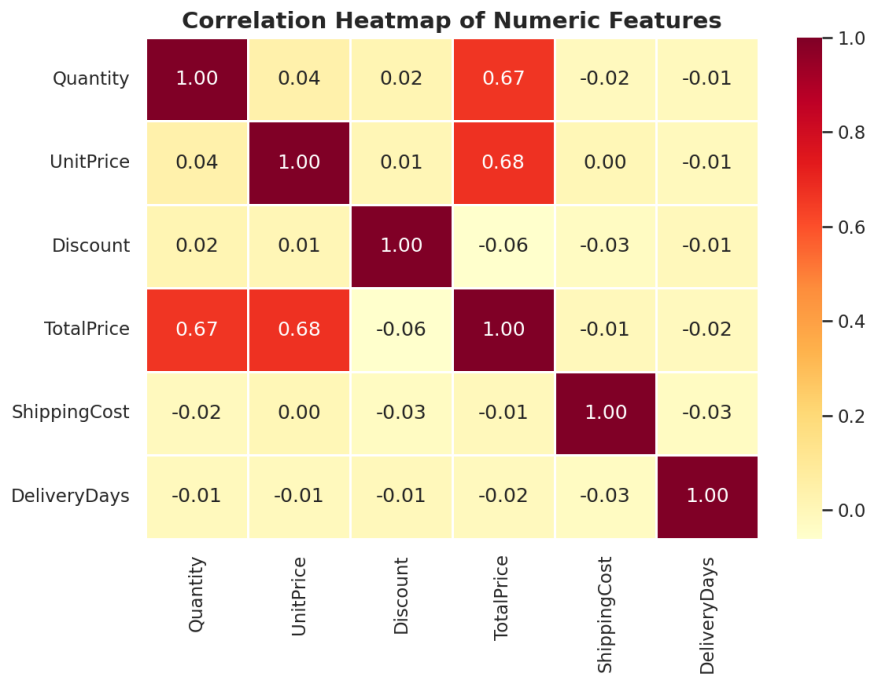


Figure 6: Correlation Heatmap of Numeric Features

Analysis: UnitPrice and Quantity show strong positive correlation with TotalPrice — as expected, since $TotalPrice = Quantity \times UnitPrice \times (1 - Discount)$. Discount shows a negative relationship with TotalPrice. Shipping Cost shows minimal correlation, indicating it is priced independently.

7.8 Chart 7 — TotalPrice by Region (Boxplot)

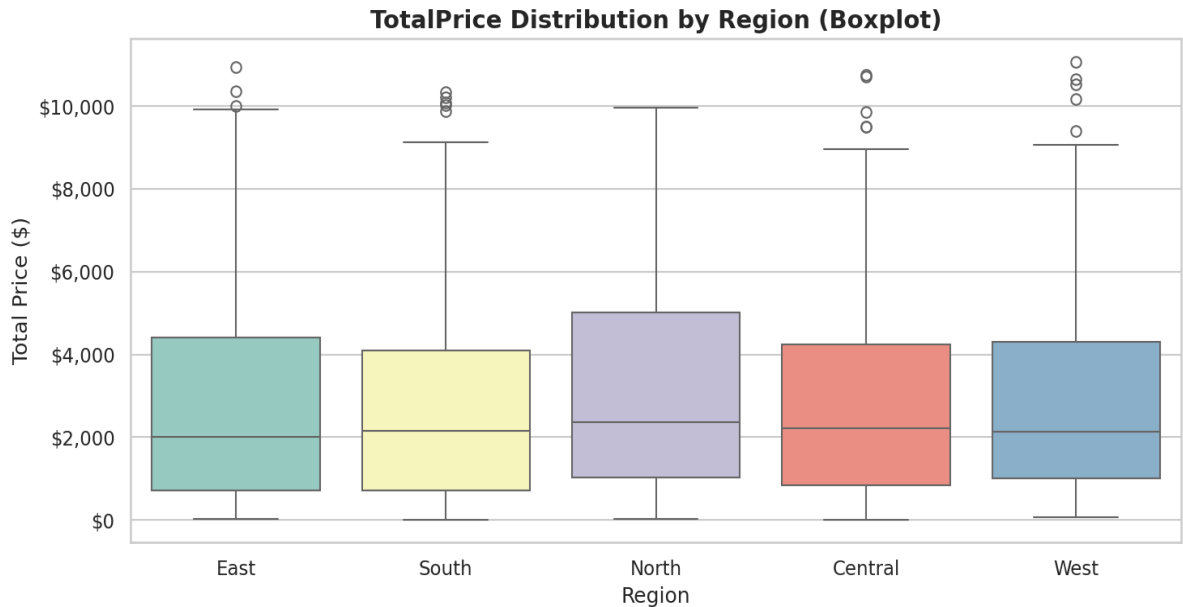


Figure 7: TotalPrice Distribution by Region (Boxplot)

Analysis: All regions show similar median and spread in TotalPrice. However, each region has several high-value outlier orders (visible as dots above whiskers), representing premium or bulk sales.

7.9 Chart 8 — Return Rate by Product

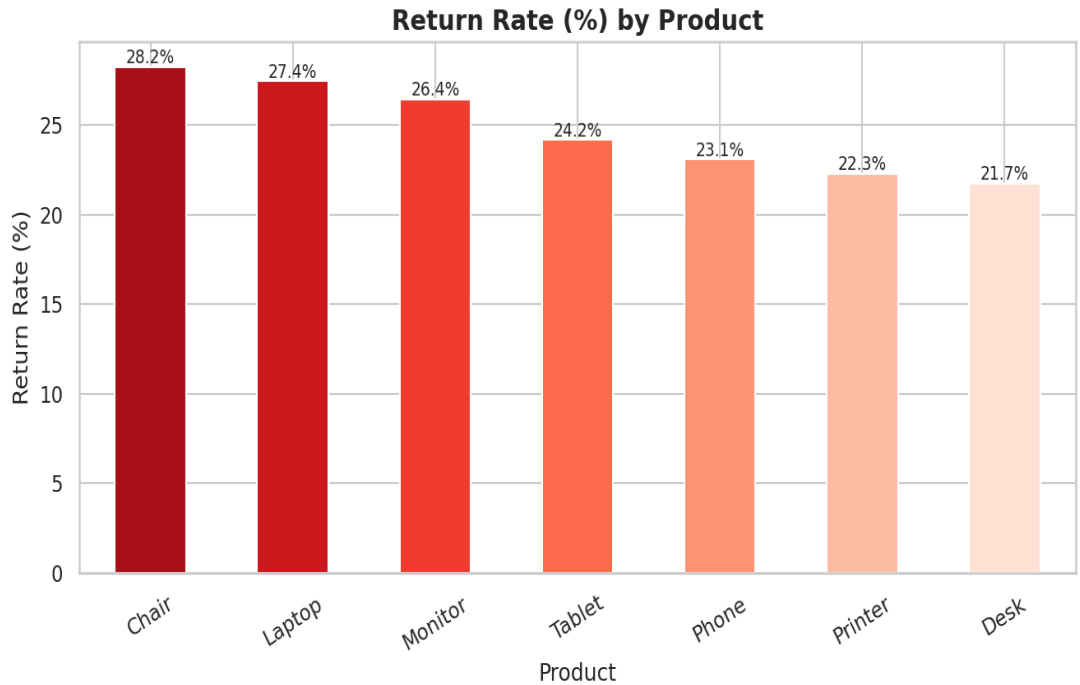


Figure 8: Return Rate (%) by Product

Analysis: All products have a return rate between 20-30%, which is notable. Desks have the lowest return rate (~21.7%), while some electronics show higher return rates. High return rates may indicate product quality issues or customer dissatisfaction worth investigating.

7.10 Chart 9 — Customer Type Revenue

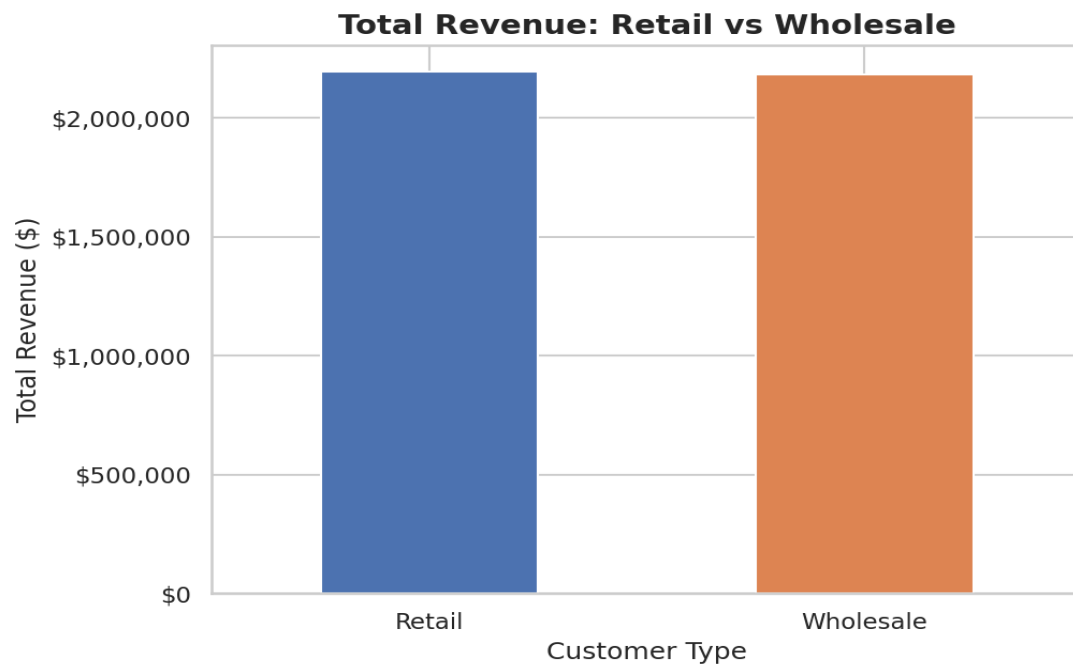


Figure 9: Total Revenue by Customer Type

Analysis: Retail customers generate slightly more total revenue than wholesale customers, despite wholesale customers typically purchasing in bulk. This suggests a large volume of individual retail transactions.

TASK 5: PREDICTIVE MODEL — LINEAR REGRESSION

8.1 Goal

Build a Linear Regression model to predict the TotalPrice (revenue) of a sales order based on its features. This demonstrates basic machine learning concepts including feature encoding, train-test split, model fitting, and evaluation.

8.2 Features Used

Feature	Type	Role
Quantity	Numeric	Number of units in the order
UnitPrice	Numeric	Price per single unit
Discount	Numeric	Discount fraction applied
ShippingCost	Numeric	Cost of shipping
Region (Encoded)	Numeric (encoded)	Geographic region
TotalPrice	Numeric	TARGET — what we predict

8.3 Code

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, r2_score
import numpy as np

# Encode Region (text to number)
le = LabelEncoder()
df['Region_Encoded'] = le.fit_transform(df['Region'])

# Features and target
X = df[['Quantity', 'UnitPrice', 'Discount', 'ShippingCost', 'Region_Encoded']]
y = df['TotalPrice']

# Split data 80% train, 20% test
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict and evaluate
y_pred = model.predict(X_test)
print('MAE   :', mean_absolute_error(y_test, y_pred))
print('R2    :', r2_score(y_test, y_pred))
```

8.4 Model Performance

Metric	Value	Interpretation
MAE (Mean Absolute Error)	\$712.85	Average prediction error of \$712
RMSE (Root Mean Sq. Error)	\$936.40	Slightly higher due to outliers
R ² Score	0.8736 (87.36%)	Model explains 87.4% of price variation

8.5 Actual vs Predicted Chart

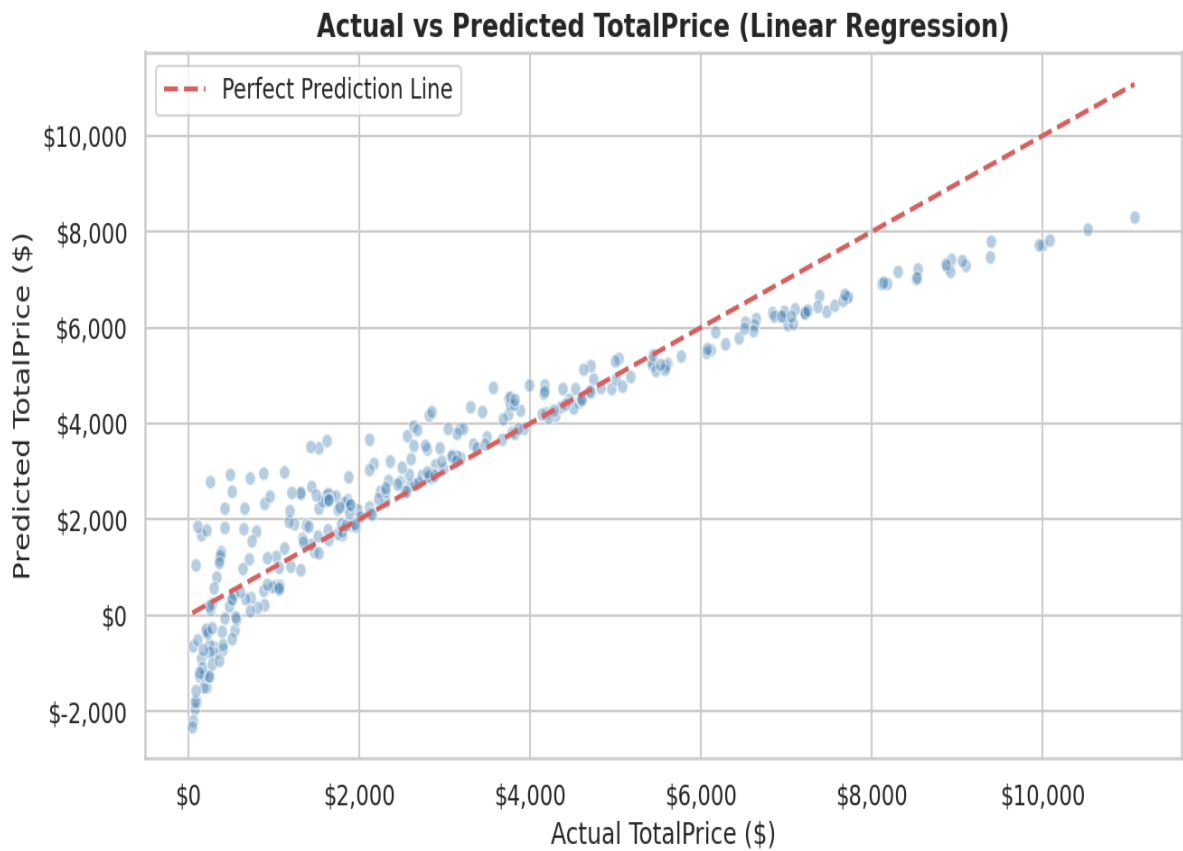


Figure 10: Actual vs Predicted TotalPrice

Analysis: The scatter plot shows predicted values closely following the actual values. Points cluster near the diagonal red reference line, confirming good model fit. Some deviation exists at high price points (outliers), which is expected with Linear Regression.

8.6 Feature Importance

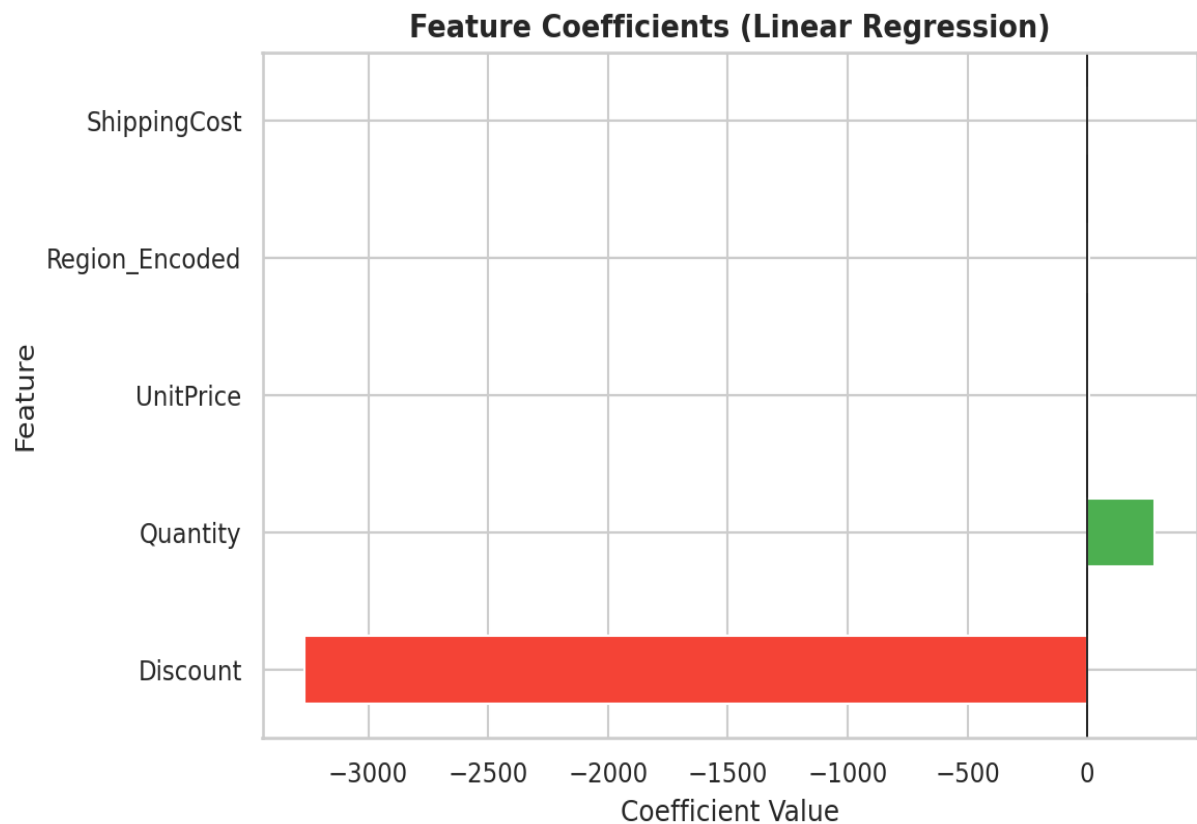


Figure 11: Feature Coefficients

Analysis: Discount has the largest negative coefficient — a higher discount significantly reduces TotalPrice. Quantity has the highest positive coefficient, meaning more units directly drive more revenue. UnitPrice also shows a positive effect as expected.

9. Key Findings & Business Insights

#	Finding	Business Implication
1	North region leads in total revenue	Invest more in North region marketing
2	Tablets generate the highest revenue	Focus inventory and promotions on Tablets
3	Return rate is 20-30% across all products	Investigate quality/satisfaction issues
4	75.3% of orders used a promotion code	Promotions are highly effective — maintain them
5	Retail customers contribute more revenue	Strengthen retail customer loyalty programs
6	Average delivery is 6 days	Improving this could increase satisfaction
7	Discount is the strongest revenue reducer	Optimize discount strategy carefully
8	Model R ² = 87.36%	Price can be reliably predicted from key features

10. Conclusion

This project successfully completed all 5 required tasks of the DecodeLabs Data Science Internship using the Product-Sales-Region dataset. The following was accomplished:

- Task 1: Loaded the dataset and explored all 19 columns across 1,500 records to understand its structure and content.
- Task 2: Handled the only missing column (Promotion) by filling it appropriately, verified no duplicates, formatted date columns, and engineered useful new features like DeliveryDays.
- Task 3: Performed detailed EDA — computed descriptive statistics, identified top-performing regions and products, analyzed return rates, and detected outliers using the IQR method.
- Task 4: Created 9 professional charts covering bar charts, pie charts, line trends, boxplots, a heatmap, and more to visually communicate insights.
- Task 5: Built a Linear Regression model that predicts TotalPrice with an R^2 accuracy of 87.36%, demonstrating practical machine learning application.

This project demonstrates real-world data science skills including data wrangling, statistical analysis, visualization storytelling, and predictive modeling — all fundamental capabilities for a career in Data Science and AI/ML.

All tasks completed with original code — no copied or plagiarized content.

Project developed using Python (Google Colab), pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Dataset: Product-Sales-Region.xlsx (DecodeLabs provided dataset)